

Student Name: _____ **PC #:** _____

ECCS 4441 – Ethical Hacking and Penetration Testing

PROJECT 1: SETTING UP PEN TESTING LAB AND UNDERSTANDING VIRTUAL ENVIRONMENT

PROJECT OVERVIEW

In this project, students set up a penetration testing lab using VMware Workstation Pro by installing and configuring virtual machines for Kali Linux, Windows 7, and Metasploitable 2. This environment provides a controlled platform for practicing and analyzing penetration testing techniques.

LEARNING OBJECTIVES

By completing this project, you will be able to:

- Explain the purpose and role of virtualization in penetration testing environments
- Install and configure virtual machines using VMware Workstation Pro
- Differentiate between attacker and target systems within a penetration testing lab
- Apply virtualization and networking concepts to create an isolated testing environment
- Demonstrate readiness for conducting basic penetration testing and vulnerability assessments

FINAL REPORT

Report Requirements

Prepare a concise report for this project, as you did most of the work in setting up and understanding the VMs and Virtual Environment. Your report must be organized, clear, and the results should be supported by evidence (screenshots).

Required Sections

1. Length: 1 – 2 pages
2. List the VMs that you installed
3. How are these VMs connected with each other in the Virtual Environment you set up.
4. List the IP addresses of all VMs
5. Write a short paragraph on Metasploitable 2

SUBMISSION REQUIREMENTS

- Final report (PDF format), upload to Canvas.
- Deadline: Tentatively February 9th, 2026.
- One Screenshot of each VMs showing IP address, include in report